# Reference, task policy and actuator realization



roadside operating context

**Reference generator**
verge line + cut height

**Vehicle layer**
constant road speed

**DLS teacher**
privileged; training only

distillation / identification

**TASK POLICY**

terrain profile · task error · state · m

**66-D input** → **MLP** hidden → **MLP** hidden → **4-D action** desired motion

**LEARNED ACTUATOR REALIZATION — one lane active per experiment**

**PPO–ROM**

**Command-conditioned ROM**
GRU + hydraulic update

**PPO–ID–FD**

**ID**
motion → valve

**FD**
valve → state

**PPO–(AAM)–ID–FD**

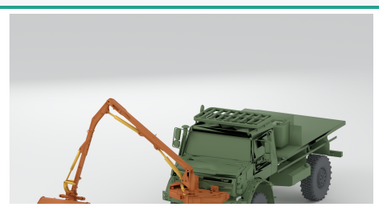
**ID**
frozen

**AAM**
zhat

**adapter**
bounded

**FD**
frozen

**Bounded PhysX target**
same for all stacks



USD / PhysX plant

measured cylinder state and actuator-model error